

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

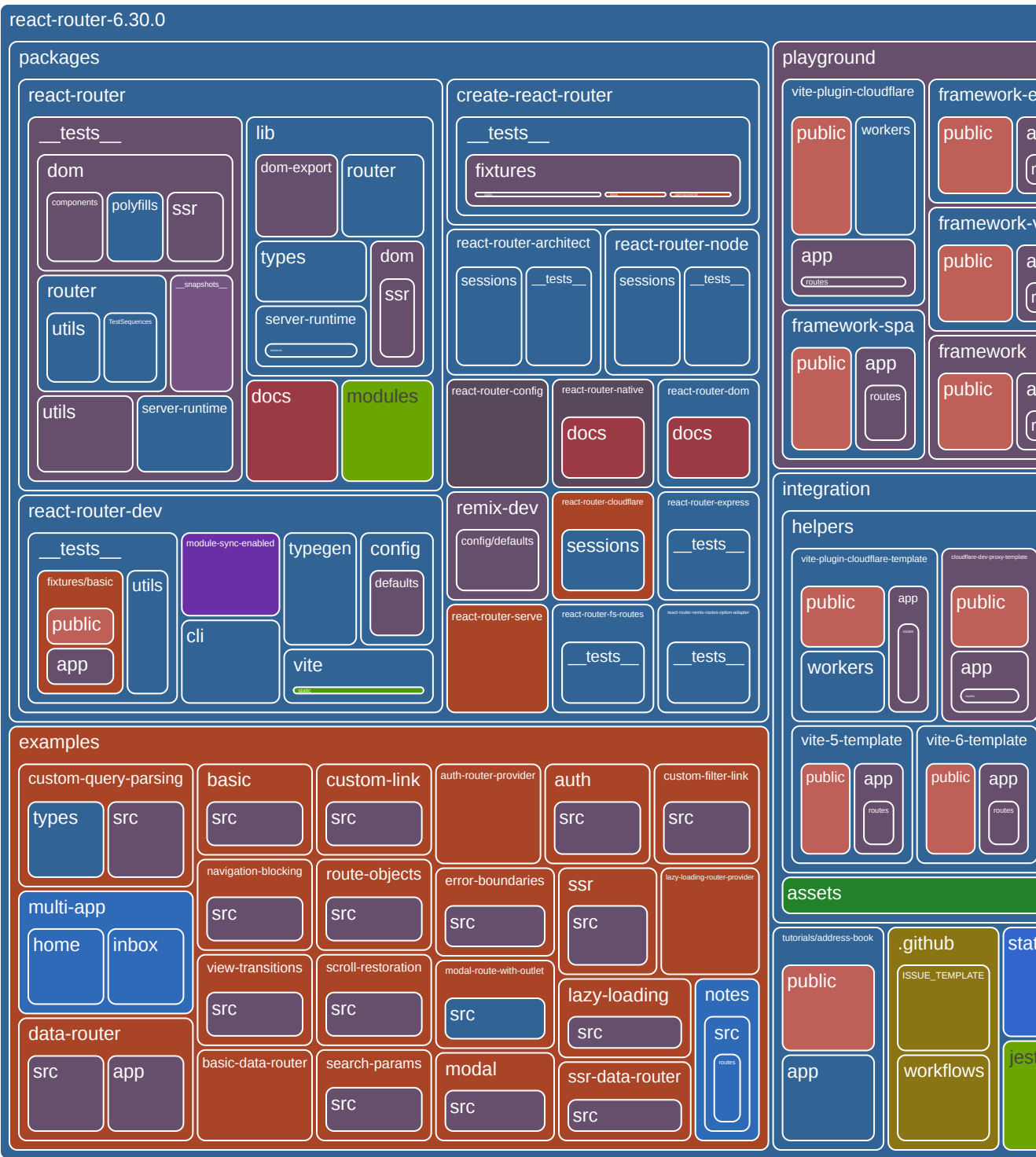
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	199.000000	199.000000	199.000000	199.000000	199.000000	199.000000
mean	22.437186	13.396985	656.366834	110.537688	392.185930	175.924623
std	100.447860	39.048509	3241.925924	221.160934	470.073657	302.739878
min	1.000000	2.000000	2.000000	2.000000	4.000000	1.000000
25%	2.000000	3.000000	22.500000	13.000000	66.000000	12.000000
50%	5.000000	6.000000	64.000000	13.000000	144.000000	27.000000
75%	13.000000	10.000000	255.000000	122.000000	616.500000	145.000000
max	1280.000000	483.000000	37507.000000	1237.000000	1756.000000	1290.000000

	directoryPath	directoryParentPath	directoryName	fileCount	firs
0	react-router-6.30.0/examples/custom-query-parsing/types	react-router-6.30.0/examples/custom-query-parsing	types	1	react-router-6.30.0/examples/custom-query-parsing/types/jsur
1	react-router-6.30.0/examples/data-router/src	react-router-6.30.0/examples/data-router	src	1	react-router-6.30.0/examples/data-router/src/la
2	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app	routes	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes/_inde
3	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template	public	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public/favico
4	react-router-6.30.0/integration/helpers/vite-5-template/app/routes	react-router-6.30.0/integration/helpers/vite-5-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-5-template/app/routes/_inde
5	react-router-6.30.0/integration/helpers/vite-5-template/public	react-router-6.30.0/integration/helpers/vite-5-template	public	1	react-router-6.30.0/integration/helpers/vite-5-template/public/favico
6	react-router-6.30.0/integration/helpers/vite-6-template/app/routes	react-router-6.30.0/integration/helpers/vite-6-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-6-template/app/routes/_inde
7	react-router-6.30.0/integration/helpers/vite-6-template/public	react-router-6.30.0/integration/helpers/vite-6-template	public	1	react-router-6.30.0/integration/helpers/vite-6-template/public/favico
8	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes/_inde
9	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	public	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public/favico
10	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	workers	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers/a
11	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes/hom
12	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public/favico
13	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	blank	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank/package
14	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	with-ignored-dir	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir/package
15	react-router-6.30.0/packages/react-router-architect/sessions	react-router-6.30.0/packages/react-router-architect	sessions	1	react-router-6.30.0/packages/react-router-architect/sessions/arcTableSessionStora
16	react-router-6.30.0/packages/react-router-cloudflare/sessions	react-router-6.30.0/packages/react-router-cloudflare	sessions	1	react-router-6.30.0/packages/react-router-cloudflare/sessions/workersKVStora
17	react-router-6.30.0/packages/react-router-config	react-router-6.30.0/packages	react-router-config	1	react-router-6.30.0/packages/react-router-config/LIC
18	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes/_inde
19	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public/favico
20	react-router-6.30.0/packages/react-router-dev/vite/static	react-router-6.30.0/packages/react-router-dev/vite	static	1	react-router-6.30.0/packages/react-router-dev/vite/static/refresh-uti
21	react-router-6.30.0/packages/react-router-dom/docs	react-router-6.30.0/packages/react-router-dom	docs	1	react-router-6.30.0/packages/react-router-dom/docs/README
22	react-router-6.30.0/packages/react-router-express/__tests__	react-router-6.30.0/packages/react-router-express	__tests__	1	react-router-6.30.0/packages/react-router-express/__tests__/server-ti
23	react-router-6.30.0/packages/react-router-native/docs	react-router-6.30.0/packages/react-router-native	docs	1	react-router-6.30.0/packages/react-router-native/docs/README
24	react-router-6.30.0/packages/react-router-node/sessions	react-router-6.30.0/packages/react-router-node	sessions	1	react-router-6.30.0/packages/react-router-node/sessions/fileStora
25	react-router-6.30.0/packages/react-router/__tests__/__snapshots__	react-router-6.30.0/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.0/packages/react-router/__tests__/__snapshots__/route-matc test.tsx
26	react-router-6.30.0/packages/react-router/__tests__/dom/components	react-router-6.30.0/packages/react-router/__tests__/dom	components	1	react-router-6.30.0/packages/react-router/__tests__/dom/components/LazyComponer
27	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills	react-router-6.30.0/packages/react-router/__tests__/dom	polyfills	1	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills/drop-Forml submit
28	react-router-6.30.0/packages/react-router/docs	react-router-6.30.0/packages/react-router	docs	1	react-router-6.30.0/packages/react-router/docs/README

Most frequent file extension per directory

Most frequent file extension per directory

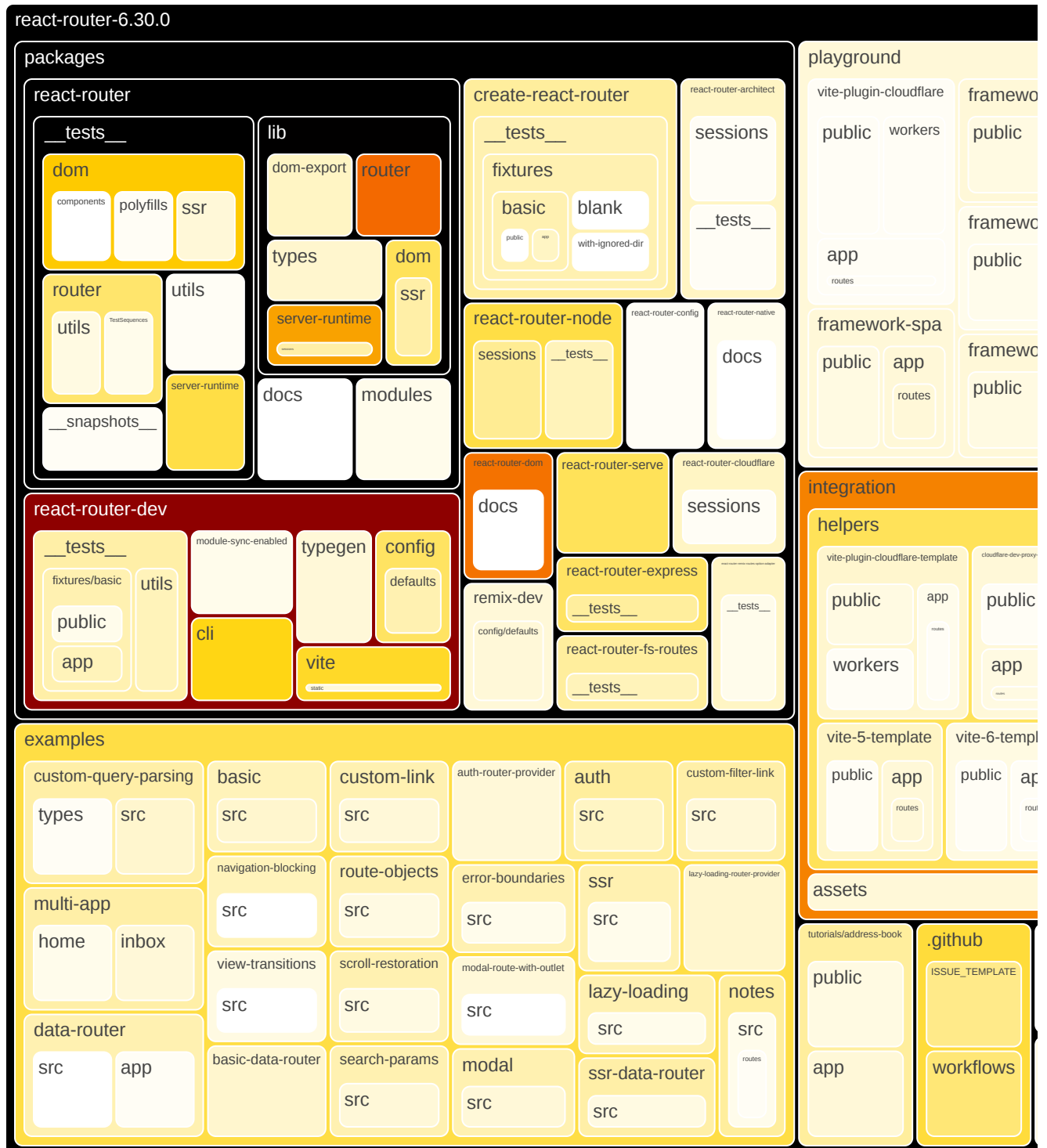


Number of commits per directory

Number of git commits

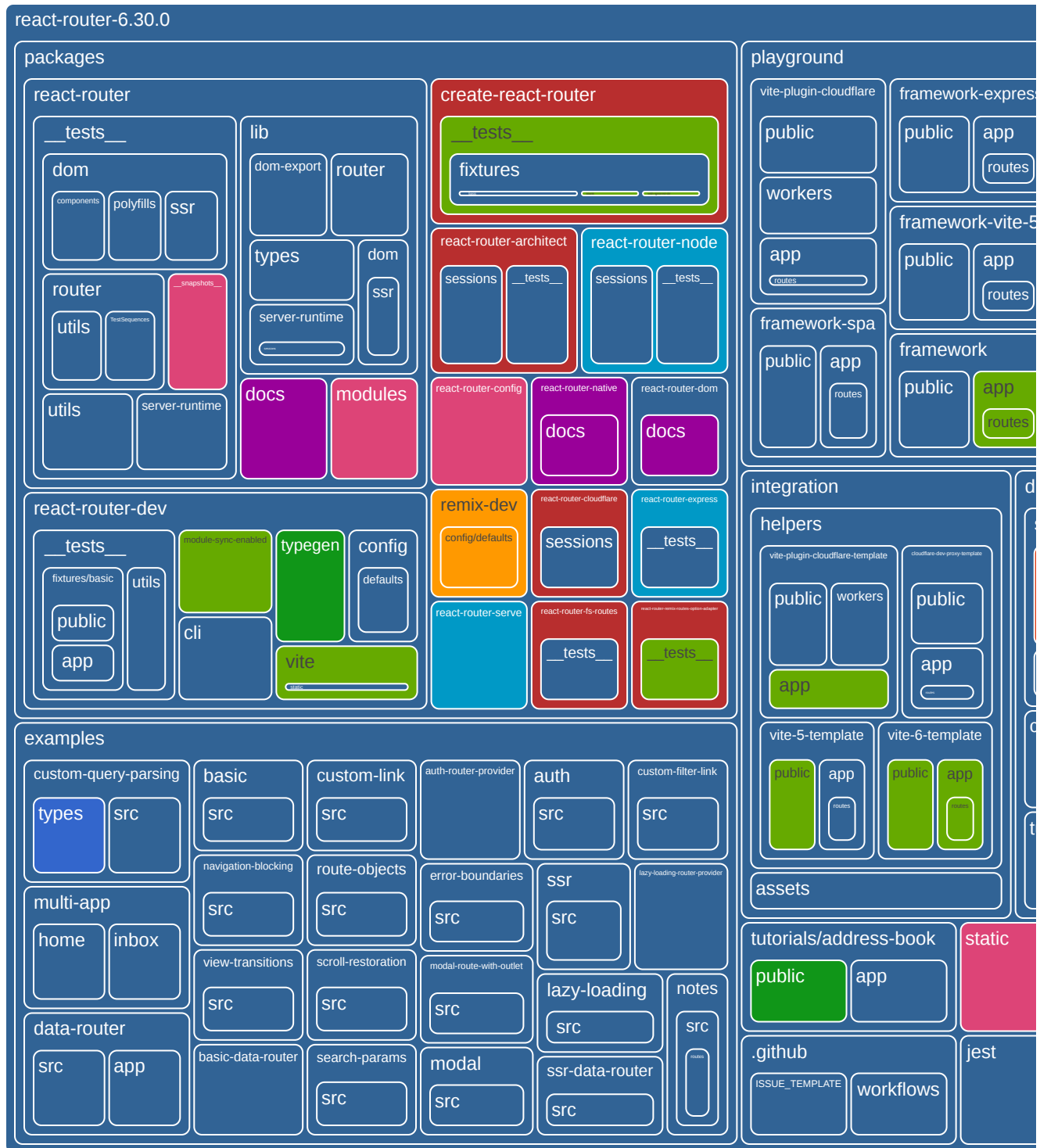


Number of distinct commit authors



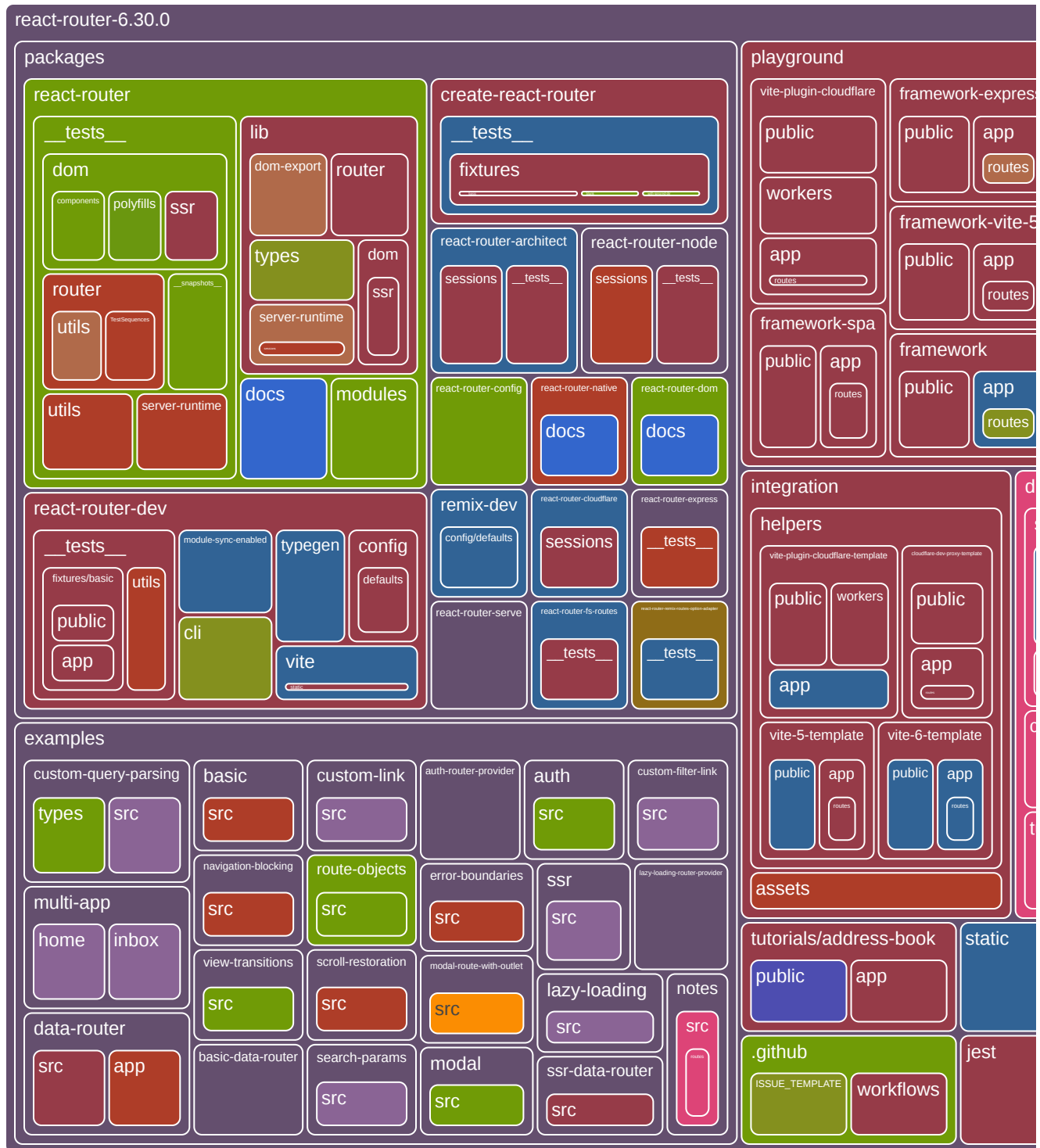
Main author per directory

Main authors with highest number of commits



Second author per directory

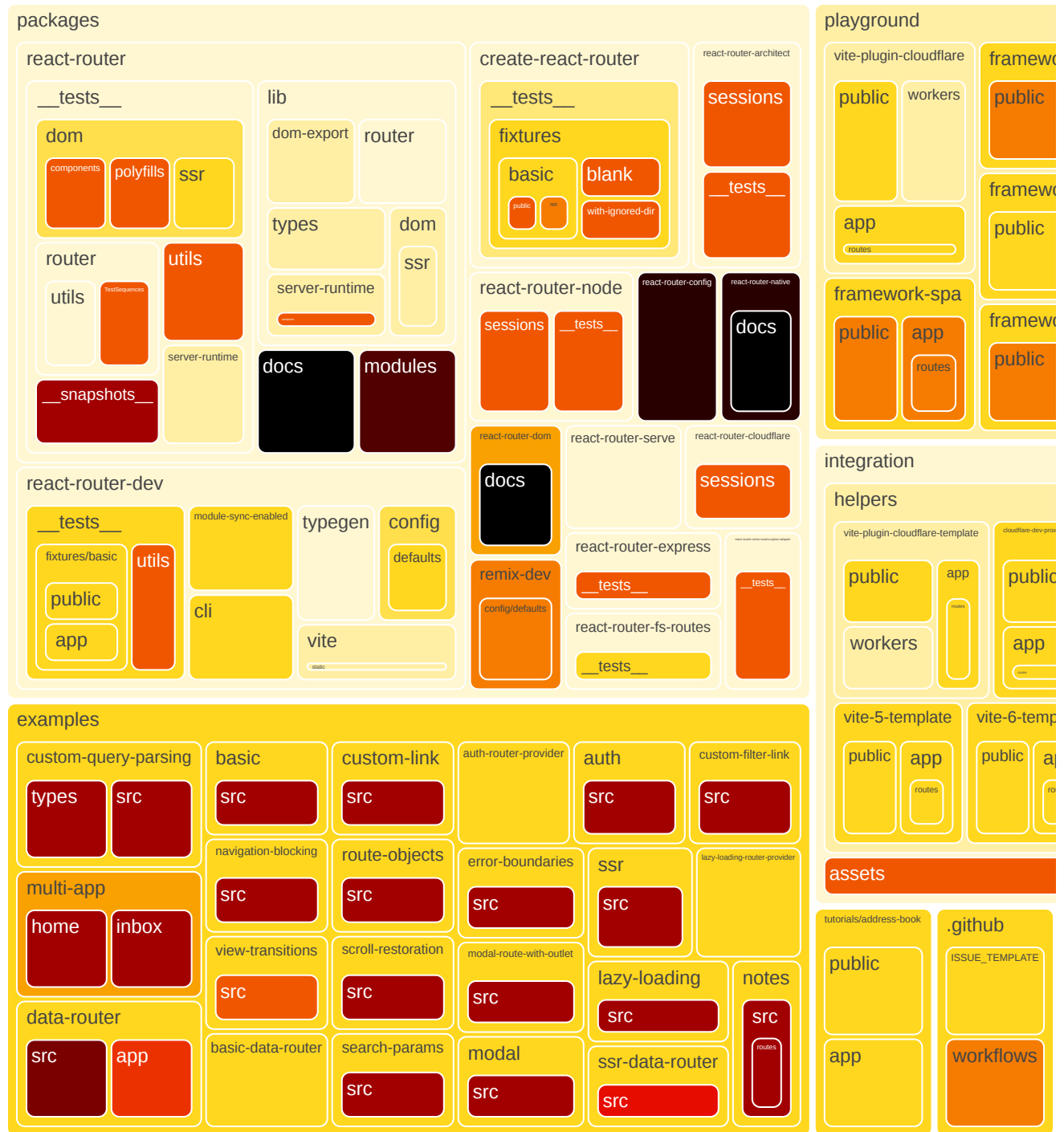
Second author with the second highest number of commits



Days since last commit per directory

Rank of days since last commit

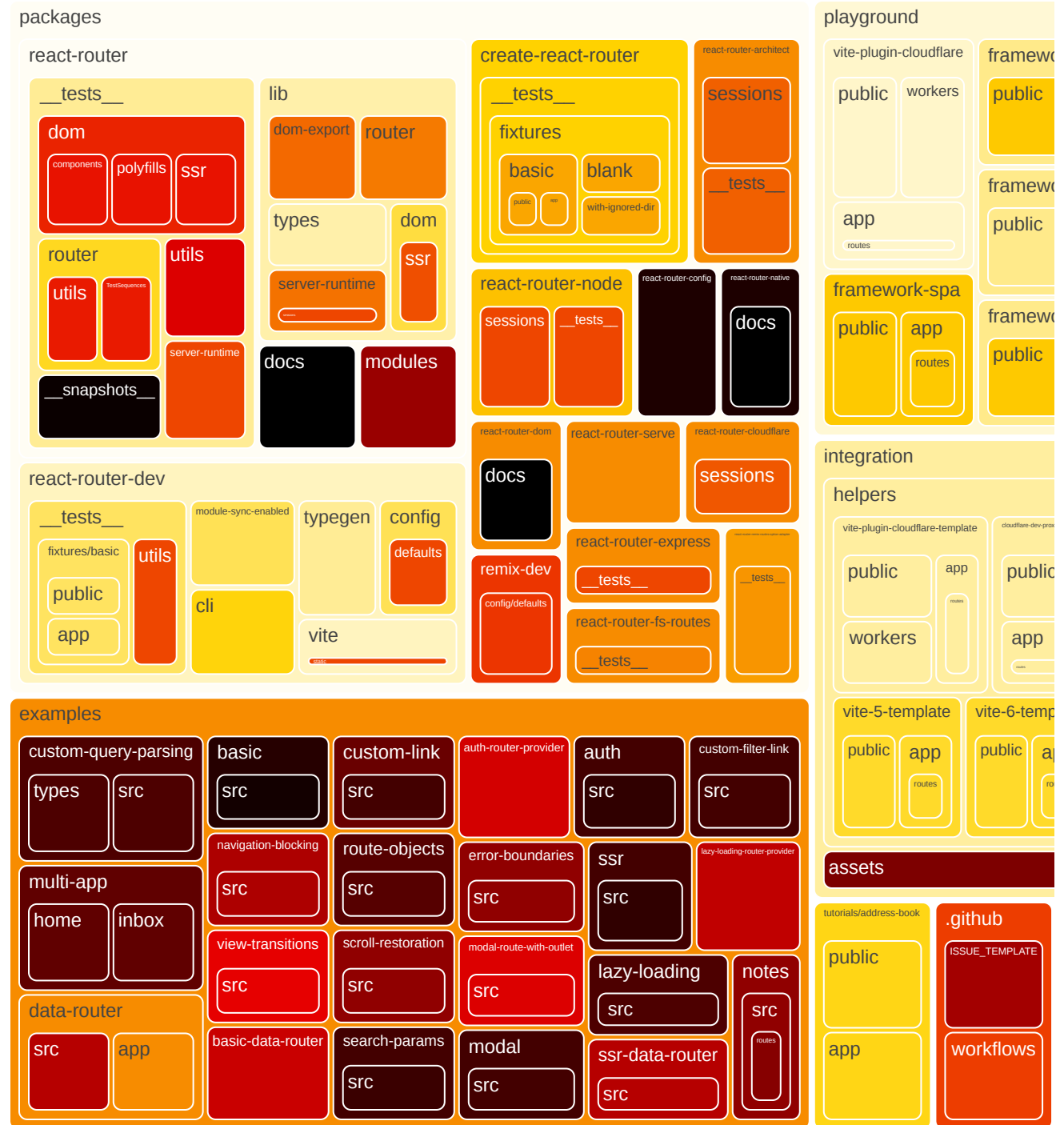
react-router-6.30.0



Days since last file creation per directory

Rank of days since last file creation

react-router-6.30.0



Days since last file modification per directory

Rank of days since last file modification

react-router-6.30.0



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

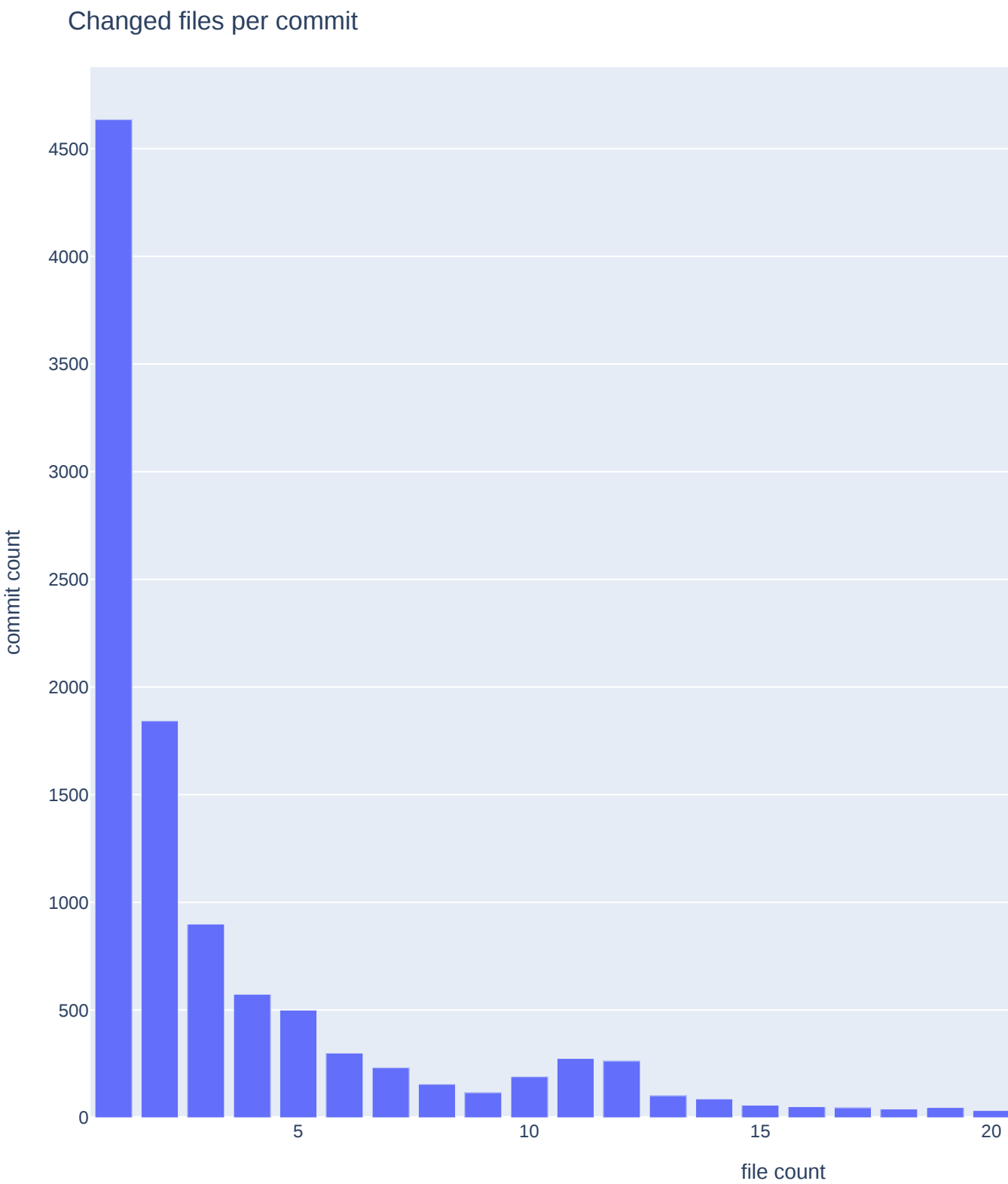
Preview data

Sum of commits that changed more than 30 files (each) = 462
Max changed files with one commit = 1061

	filesPerCommit	commitCount
count	171.000000	171.000000
mean	149.959064	65.777778
std	193.597592	390.563903
min	1.000000	1.000000
25%	43.500000	1.000000
50%	87.000000	3.000000
75%	157.500000	8.000000
max	1061.000000	4635.000000

	filesPerCommit	commitCount
0	1	4635
1	2	1842
2	3	899
3	4	573
4	5	499
5	6	300
6	7	232
7	8	155
8	9	116
9	10	190
10	11	275
11	12	263
12	13	102
13	14	87
14	15	58
15	16	51
16	17	46
17	18	40
18	19	47
19	20	33
20	21	36
21	22	31
22	23	53
23	24	30
24	25	30
25	26	15
26	27	56
27	28	56
28	29	19
29	30	17

Bar chart with the number of files per commit distribution



WordCloud of git authors

	word	frequency
0	Matt Brophy	2059
1	Michael Jackson	1951
2	Ryan Florence	1306
3	Chance Strickland	511
4	Remix Run Bot	493
5	Mark Dalgleish	418
6	Pedro Cattori	412
7	Tim Dorr	390
8	Jimmy Jia	389
9	Jacob Ebey	290

